

Venkateswara Sahu

Applied AI and Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Open to relocation

PROFILE

2026 Computer Science graduate with LPU-industry Generative AI internship experience. Built Text-to-SQL and document-processing workflows, and published a Python package for drift monitoring.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++

Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, Optuna

Generative AI: LangGraph, LangChain, RAG, FAISS, sentence-transformers, LLM APIs

Engineering: FastAPI, Flask, Docker, GitHub Actions, Kafka, Airflow, MLflow, TiDB Cloud

EXPERIENCE

TransOrg Analytics (Pickl.AI) - LPU Industry Tie-Up

Punjab, India

Generative AI Intern

Jan 2026 – May 2026

- Led the technical implementation of F1InsightAI, a nine-node LangGraph pipeline that translated natural-language questions into SQL across 700,000+ Formula 1 records in 14 TiDB tables.
- Implemented FAISS schema retrieval, conditional routing, SQL validation, and error-guided retries; refined retrieval using enriched schema descriptions and table co-occurrence rules.
- Re-evaluated in Oct 2026: matched independent reference results on 39/40 first attempts (97.5%); retrieval MRR@7 0.678 to 0.888 on the same frozen snapshot.
- Exposed the workflow through a Dockerized Flask API and a public [Hugging Face demo](#) with query traces and retrieval diagnostics.

PROJECTS

Vigil (vigil-drift) - Drift Monitoring and Feature Attribution

Jun–Sep 2026 | [GitHub](#) | [PyPI](#)

Python, PyTorch, Kafka, Airflow, MLflow, FastAPI

- Published a Python package for label-free drift monitoring using adaptive and frozen autoencoders, reconstruction-error tests, and kernel density estimation.
- Compared nine feature-attribution ranking methods across 20 seeds, 18 synthetic conditions, and six controlled CICIDS2017 development-data conditions; documented strengths and failure modes.
- Added regression tests, package installation checks, and GitHub Actions, with FastAPI, Kafka ingestion, and Airflow retraining examples.

P&ID Document Intelligence and Material Take-Off Extraction

Jul 2026 | [GitHub](#)

Python, YOLOv8, OCR, NetworkX, LangGraph

- Combined symbol detection, connected-component-guided OCR, and spatial graph matching to convert engineering drawings into structured equipment and instrumentation records.
- Reduced OCR processing time from approximately 360 seconds to 7 seconds in one project benchmark, and added rule-based validation that flags low-confidence entity relationships for review.

CTR Prediction and Candidate Ranking

2025; Oct 2026 rebuild | [GitHub](#) | [Live Demo](#)

Python, XGBoost, LightGBM, Optuna, Flask, Streamlit

- Rebuilt a 123-feature Criteo pipeline with training-only preprocessing and out-of-fold target encoding; compared logistic regression, XGBoost, and LightGBM.
- Measured **ROC AUC 0.7605** and log loss 0.4857 on **249,987 held-out rows**; served batch-stable predictions through Flask and Streamlit.

EDUCATION

Lovely Professional University

Aug 2022 – Jun 2026

B.Tech (Hons.) Computer Science and Engineering - Data Science and Data Engineering

Punjab, India

CGPA: 8.38/10